

$$Q = 2(i\epsilon)$$

$$A = 2(i)$$

i) a) 0.00072

$$7.2 \times 10^{-4} \text{ m}$$

b) 8,900,000

$$8.9 \times 10^6 \text{ m}$$

ii) 7.2×10^{-4}

$$7.2 \times 10^{-4} \text{ m}$$

$$0.72 \times 10^{-4} \times 10^1$$

$$0.72 \times 10^{-3}$$

$$0.72 (\text{milli}) \text{ mm}$$

Q: 2(ii) (or)
A: 2(ii) (or)

(a) 45.268

~~45.3~~

45.3

(b) 0.005672

0.00567

(c) 9432

9430

Q: 2(iii) (or)
A: 2(iii)

(a) distance :- 6 km + 8 km
14 km

(b) By finding displacement (magnitude)
we use by using pythagoras theorem:-

$$\begin{aligned} & \sqrt{(6)^2 + (8)^2} \\ &= \sqrt{36 + 64} \\ &= \sqrt{100} \end{aligned}$$

$$= 10 \text{ km}$$

{ North-east of the
starting point }

Q: 2 (iv)
A: 2 (iv)

① Acceleration:

$$a = \frac{\Delta v}{t}$$

$$= \frac{v_f - v_i}{t} \Rightarrow \frac{25 - 5}{8}$$

$$= \frac{20 \text{ m/s}}{8 \text{ s}} = \frac{5}{2}$$

$$= 2.5 \text{ m/s}$$

② velocity :-

$$\vec{v} = \frac{\Delta v}{\Delta t}$$

$$= \frac{v_i + v_f}{2} \Rightarrow \frac{5 + 25}{2}$$

$$= \frac{30}{2} \Rightarrow 15 \text{ m/s}$$

③ distance :- $s = vt$

$$15 \text{ m/s} \times 8$$

$$120 \text{ m/s}^2$$

$$120 \text{ m}$$

Q.2(v)
A:2(v)
According to ~~se~~ Newton's second Law :-
 $F = ma$

Switch up to find acceleration:-

$$a = \frac{F}{m}$$

①

(force :- 18N, mass: 4kg)

$$a = \frac{18}{4}$$

$$a = 4.5 \text{ m/s}^2$$

②

$$a = \frac{F}{m}$$

(same force :- $m = 12 \text{ kg}$)

$$= \frac{18}{12}$$

$$= 1.5 \text{ m/s}^2$$

ⓐ Second newton law :-

The ~~acce~~ acceleration of an object is directly proportional to the magnitude of net force, in a same direction, is inversely proportional to the mass of an object.

$$Q: 2(v_i)$$

$$A: 2(v_i)$$

Impulse := change in momentum

$$= m(v_f - v_i)$$

$$= 0.2(0 - 15 \text{ m/s})$$

$$= 0.2(-15)$$

$$= \cancel{-3 \text{ kg} \cdot \text{m/s}}$$

$$= -3 \text{ kg} \cdot \text{m/s} \quad (\text{magnitude: } 3 \text{ N} \cdot \text{s})$$

Force_{average} (F) := ~~AV~~

$$= \frac{\Delta p}{\Delta t}$$

$$= \frac{-3}{0.1}$$

$$= -30 \text{ N} \quad (\text{magnitude: } 30 \text{ N})$$

The '-' negative sign shows the force acts opposite to the Ball's original direction of motion (bringing it to rest).

Q: 2 (vi) (or)

A: 2 (vii) (or)

Principle of moments:

for clockwise moments = anticlockwise moments
 $\text{Weight}_A \times \text{Distance}_A = \text{force}_B \times \text{distance}_B$

$$980\text{N} \times 0.25\text{m} = 400\text{N} \times \text{distance}_B$$

$$120\text{Nm} = 400\text{N} \times \text{distance}_B$$

$$\frac{120}{400} = \text{distance}_B$$

$$0.3$$

$$0.3 = \text{distance}_B$$

Q: 2 (viii) (or)

A: 2 (viii) (or)

$$V_{\text{avg}} = \frac{2\pi r}{T}$$

$$= \frac{2 \times 3.14 \times 7,000,000\text{m}}{6000}$$

$$= \frac{43,982,000}{6000}$$

$$= 7330\text{ m/s}$$

$$V_{\text{avg}} = 7.33\text{ km/s}$$

$$Q: 2(\text{m})$$

$$A: 2(\text{m})$$

Force stretching spring = weight = mg

$$= 3\text{kg} \times 10\text{m/s}^2 = 30\text{N}$$

By Hooke's law,

$$K = \frac{F}{x}$$

$$= \frac{30}{0.06}$$

$$K = 500\text{N/m}$$

Q: 2(m)

A: 2(m)

$$KE = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 2 \times (12)^2$$

$$= \frac{1}{2} \times 2 \times (144)$$

$$= 144$$

$$= 288$$

$$= 144\text{J}$$

If speed doubles to 24 m/s ..

$$KE_2 = \frac{1}{2} mv^2$$

$$= \frac{1}{2} (2)(24)^2$$
$$= \frac{1}{2} (2) 576$$
$$= 576 \text{ J}$$

Q: 2(ni)(or)
A: 2(ni)(or)

Biophysics:-

Studies characteristics and study of system of living things. e.g fluid dynamics of blood flow.

Astrophysics:-

Studies the physical nature of stars and celestial bodies. E.g; the formation of stars.

Acoustics:-

Studies the production, transmission and reception of sound. E.g designing concert hall acoustics.

$$Q: 3(i)(or)$$

$$A: 3(i)(or)$$

Screw gauge :-

is a device used to measure a fraction of a smallest division on a linear scale by rotating a circular scale over it, it measures lengths even smaller than a vernier caliper.

Pitch :-

is the distance moved by the circular scale along the main scale in one complete rotation.

Least count :- $\frac{\text{pitch}}{\text{divisions}}$

$$= \frac{1 \text{ mm}}{100} = 0.01 \text{ mm}$$

Diameter calculation :-

Diameter :- main scale reading + (circular scale reading $\times$ least count)

$$4 + (32 \times 0.01)$$

$$4 + 0.32$$

$$= 4.32 \text{ mm}$$

Q.3 (ii) (or)
A.3 (ii) (or)

Examples :-

first law :-

A book on a table stays at rest until a force moves it.

2nd Law :-

Pushing a shopping trolley harder produces greater acceleration.

3rd law :-

A swimmer pushes water back and is pushed forward in reaction.

Conservation of momentum setup :-

$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v$, since the car is initially at rest.

$$(v_2 = 0)$$

Calculation :-

$$(2000 \times 15) + (800 \times 0) = (2000 + 800)$$
$$= 30,000 + 0 = 2800$$

$$\Rightarrow \frac{30,000}{2800}$$

$$= 10.71 \text{ m/s}$$

Q: 3 (iii) (or) _____
A: 3 (iii) (or) _____

Definition :-

centripetal force :-

is the force that pulls an object out of its straight-line path and into a circular path that is directed through towards the centre.

Orbital motion :-

is the region regular, repeating circular (or near circular) path that a satellite follows around a larger body due to gravity acting as the centripetal force.

② Average orbital speed :-

$$v = \frac{2\pi r}{t}$$

$$= \frac{2 \times 3.14 \times 8,000,000}{7200}$$

$$= \frac{50,265,000}{7200}$$

$$= 6981 \text{ m/s}$$

② Centrifugal force :-

$$F_c = \frac{mv^2}{r}$$

$$= \frac{500 \times (6981)^2}{8,000,000}$$

$$= \frac{500 \times 48,738,000}{8,000,000}$$

$$\approx \cancel{3000} \quad 3046 \text{ N}$$

Q: 3 (iv) (or)
A: 3 (iv) (or)

derivation :-

A column of liquid of height (h)
; base Area (A) ; density (ρ)

The force on the base equals
the weight of the liquid
column :-

$$F = W = mg \quad \left\{ \begin{array}{l} \text{since } \text{density } \rho = \frac{m}{V} \end{array} \right.$$

and $V = Ah$, mass $m = \rho Ah$ so

$F = \rho Ahg$. Since pressure $P = \frac{F}{A}$
substituting gives $P = \frac{\rho Ahg}{A}$

$$P = \rho hg$$

Numerical :-

$$P = \rho gh = 1000 \text{ kg/m}^3 \times 10 \text{ m/s}^2 \times 6 \text{ m}$$
$$60,000 \text{ Pa} = 60 \text{ kPa}$$

MCQs :-

1. A
2. B
3. C
4. D
5. C
6. C
7. B
8. A
9. ~~B~~ D
10. C
11. D
12. B